

Can a 93% Satellite AI Really See Deforestation? Testing a Benchmark Model in the Real Amazon

[Your Name] — [School, City, Country] — [Month Year]

Introduction

Deforestation is really a major problem in the human world. It causes many major problems such as flooding, increasing temperatures, habitat destruction, and many other bad things. As a solution, there are models built to detect deforestation — but are they really helpful, and will they predict correctly in real life? That is what we will find out.

For a model to be trained, we need a large dataset containing many satellite images. We will use EuroSAT — 27,000 labeled satellite images. But building a model from zero is time-wasting and expensive, and that is why we will use ResNet50, a 50-layer neural network that is already trained and ready to use. But what will I benefit from a model trained on cat and dog images? Here is the catch: a model trained on 1.2 million images already recognizes patterns, edges, textures, and small details other models can't find. So after we give it our EuroSAT dataset and test it, it is expected to give high accuracy.

Here is how it will work. The model classifies each image into one of 10 land types that we set: AnnualCrop, Forest, HerbaceousVegetation, Highway, Industrial, Pasture, PermanentCrop, Residential, River, and SeaLake. We then measure its accuracy on EuroSAT — this is the benchmark. But for the real test, we will test on Amazon imagery of the same place in 2018 vs 2024. Deforestation is detected by our own rule: any patch of land the model labels Forest in 2018 and not-Forest in 2024 gets flagged. Only after the model makes all its claims do we compare them to the Hansen Global Forest Change dataset — the official scientific record of where forest loss actually happened — and grade our model against it.

A model that scores ~93% in the lab — how much does it really score in the real world, and why does it lose the difference?

Phase 1 — Training and testing on the benchmark

First step: we load the EuroSAT dataset using PyTorch. The dataset downloaded with no problems. Now we make a small CNN class as our baseline model. The baseline gave three accuracy results: 57.0% → 62.0% → 63.2%, for epochs 1, 2, and 3 respectively. Epochs are the number of complete passes the model makes through the training set. You can't say "the more epochs the more accuracy," because it levels off — like a light intensity vs photosynthesis graph, but with a decline phase after the plateau if you keep going (overfitting).

Now we train the transfer-learned ResNet50 on the same dataset and see the results. The results in ascending epoch order are 90.0% → 91.4% → 93.0%. The model understood the data and patterns correctly — very good results compared to the baseline. Transfer learning bought us about 30 points of accuracy over the from-scratch baseline (63.2% → 93.0%): that is its measured value.

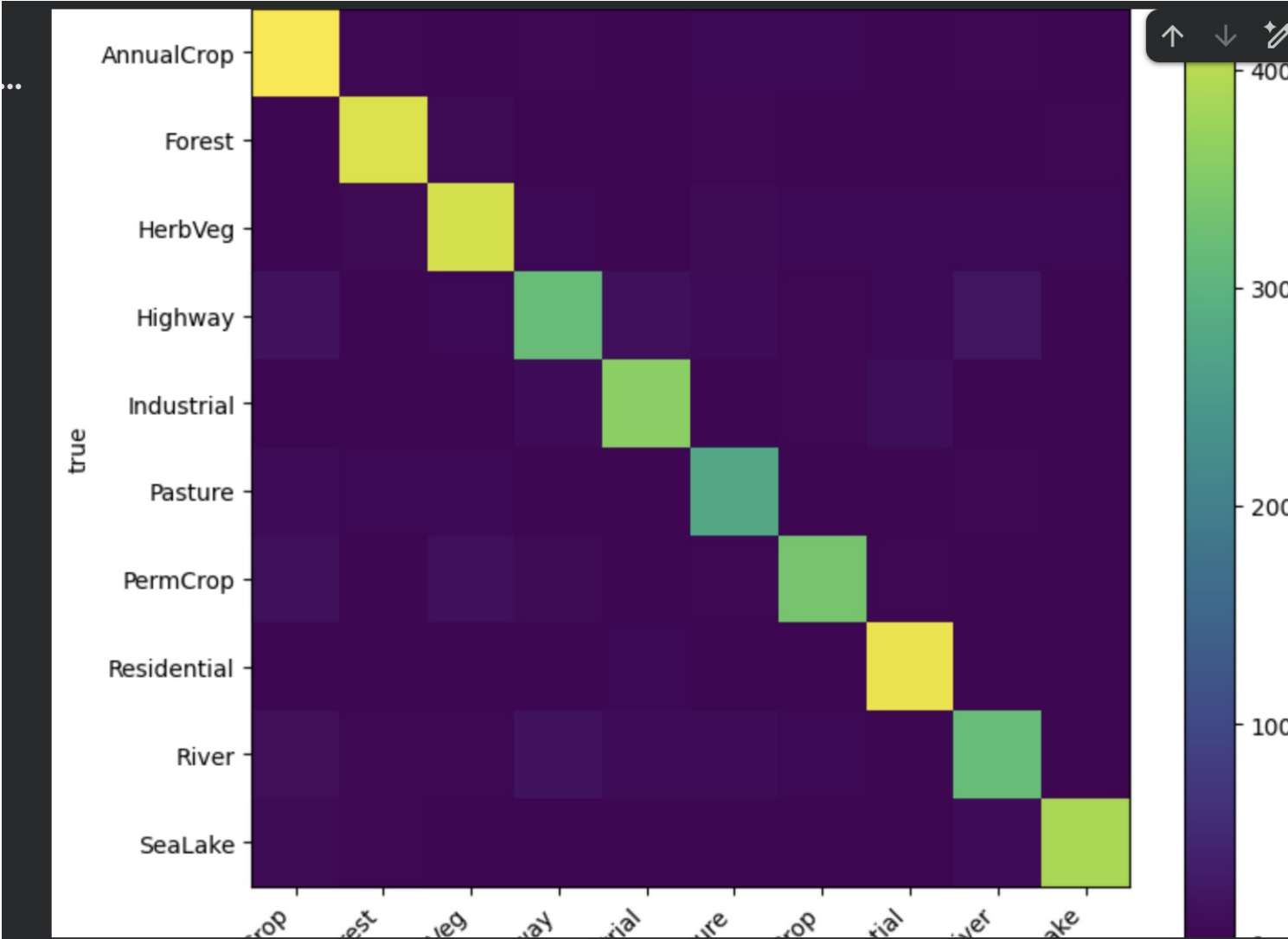


Figure 1: Confusion matrix of the ResNet50 model on the EuroSAT test set.

Figure 1 shows how the model predicted the test data. Each row is the true class and each column is the model's prediction, so the bright diagonal staircase is all the correct predictions. Bright yellow ON the diagonal means strength — but any light OFF the diagonal would mean a repeated confusion, so "more yellow" is only good in the right place. The model's weakest diagonal box still holds around 270 correct predictions, and everything off the diagonal stays close to zero. These are very good and satisfying benchmark results.

The model gave excellent results on its benchmark — but how will it predict in messy real life? That is what we go through now: we test the model on real Amazon data and compare it to Hansen. Let's get started.

Phase 2 — The real world: Rondônia, Brazil

The region we test our model on in the Amazon is Rondônia, Brazil, with the coordinates [-63.30, -9.20, -63.20, -9.10] — about an 11×11 km rectangle. We chose it because the forest loss there in 2018–2024 is so dense, documented, and undeniable that any working pipeline must detect it — if our model finds nothing here, the problem is the model, not the world.

Now we fetch the images we test the model on. The imagery is Sentinel-2 satellite data, exported through Google Earth Engine. For each year we build a cloud-filtered (<20% cloud) June–August median composite — the median combine erases clouds statistically, the same way the median beat outliers in our previous research, just applied to pixels this time. The 2018 composite drew on 12 satellite images and the 2024 composite on 28. We use the same season for both years so that a dry field next to its wet-season self doesn't look like fake deforestation.

Both images are then cut into 64×64-pixel tiles — the same size as EuroSAT's images — giving a 17×17 grid of 289 patches, where each patch covers about 640×640 meters of land. The model classifies every patch of both years, producing a land-cover map of 2018 and one of 2024.

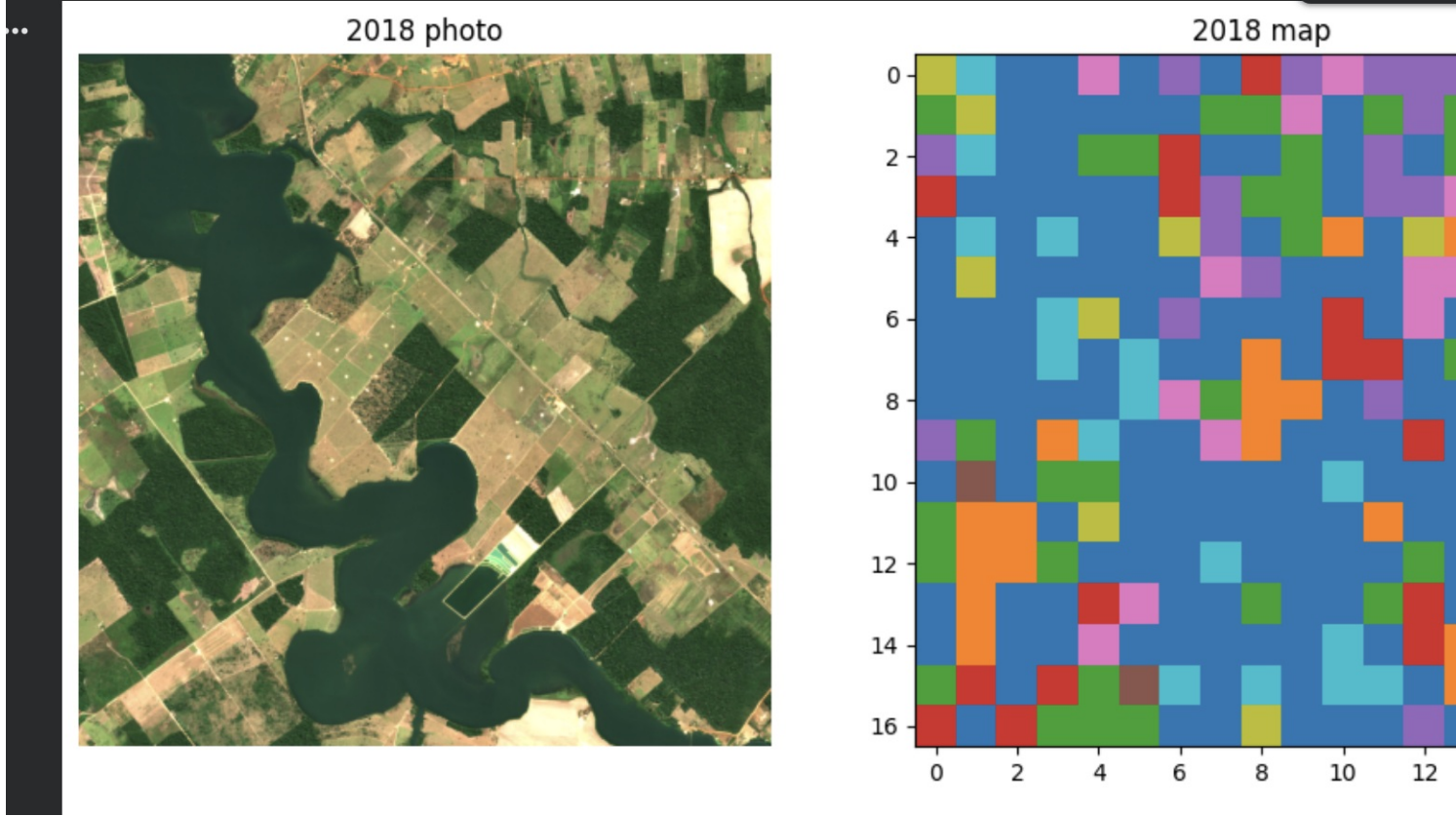


Figure 2: The 2018 Sentinel-2 composite of the region next to the model’s land-cover map of it.

The first maps immediately looked suspicious: spatially incoherent “confetti” with no connected regions, even though the photo has an obvious river system and large coherent forest blocks. The model’s Forest fraction was only 11% in 2018 and 4% in 2024 — in the middle of the Amazon.

Diagnosing the problem

Mean confidence on the Amazon 2018 image was 0.50, while on EuroSAT it was ~0.95. Mean confidence is the average of the model’s own certainty across all patches — how sure it feels about its answers. At 0.50, the model is telling us in its own arithmetic that it does not recognize this world.

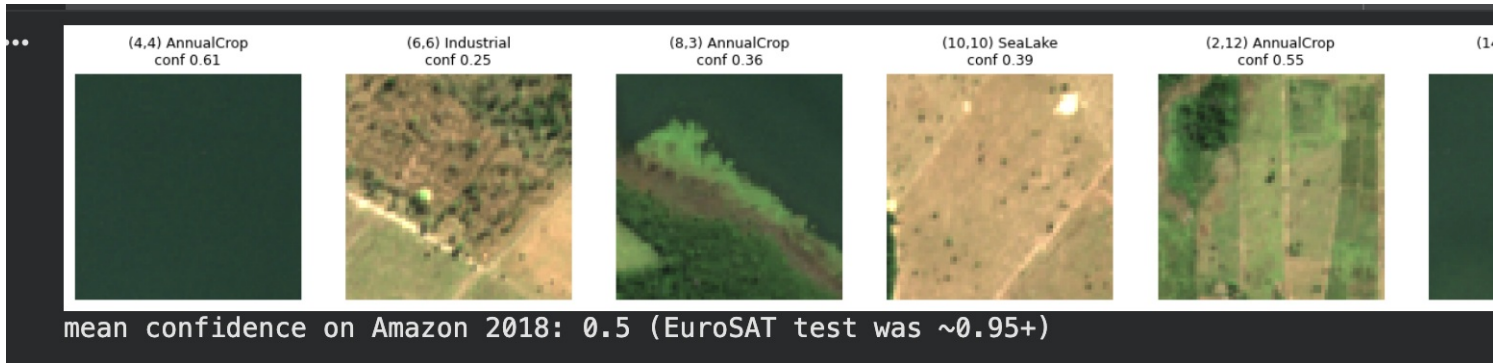


Figure 3: Six diagnostic patches with the model’s verdict and confidence — dense forest labeled AnnualCrop.

Figure 3 shows the model catching itself in the act: patches of solid dark forest canopy labeled “AnnualCrop” with confidence 0.61. After we got that 0.50 confidence, we wanted to check whether brightness has an effect, so we tested a brighter preprocessing. The confidence changed only to 0.524 — no major change, so brightness is not the cause. But something stranger happened: the same forest patches flipped their labels from AnnualCrop to SeaLake at confidence 0.83.

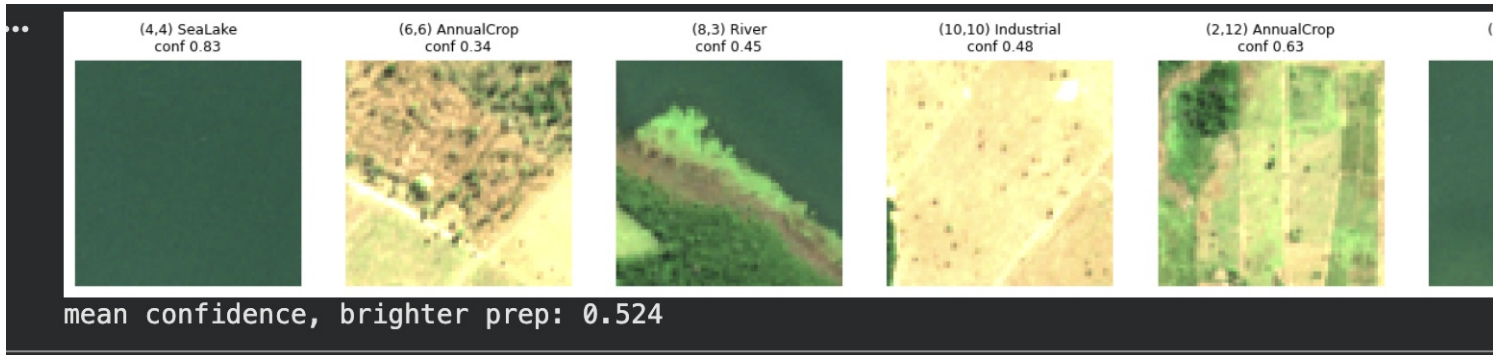


Figure 4: The same patches under brighter preprocessing — labels flip to SeaLake with higher confidence.

Two findings from this experiment: the confusion is not a brightness artifact — the domain gap is the dominant cause; and the model’s predictions are unstable under preprocessing choices, which shows that high confidence does not equal correctness. We stopped preprocessing experiments after two controlled tests and kept the original preparation.

Phase 3 — Grading the model against Hansen

Now we flag deforestation with our rule — Forest in 2018 AND not-Forest in 2024 — and finally the part the whole research was made for: testing our model against Hansen and grading it. The Hansen loss map is exported for the exact same rectangle, and a patch counts as Hansen-loss if more than 10% of its pixels show loss during 2019–2024 (loss stamped 2018 is excluded, because clearing from early 2018 is already gone in our “before” composite, so the model cannot possibly flag it).

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.. flagged patches: 20
Hansen-loss patches: 63
TP=12 FP=8 FN=51 TN=218
precision=0.60 recall=0.19
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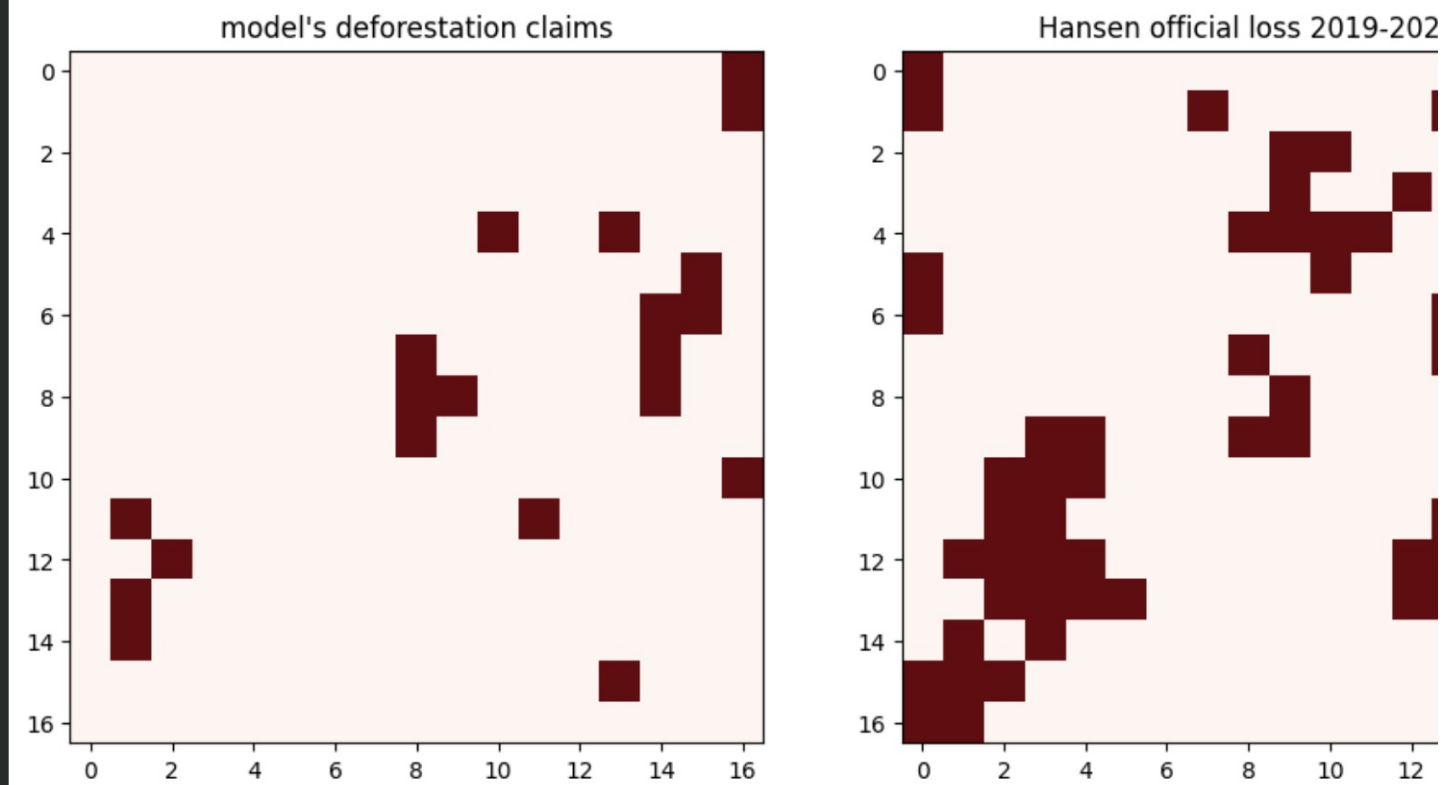


Figure 5: The model's deforestation claims next to Hansen's official loss record, with the metrics.

Here are the metrics. Our model flagged 20 patches as deforestation. Out of those 20, 12 were confirmed by Hansen — a precision of 60%. But Hansen records 63 patches of real forest loss in the same window, so recall is only 19%: the model missed 81% of the real deforestation. This disqualifies the model as a monitoring tool in this form. And it demonstrates exactly what this research was designed to measure: benchmark performance can collapse in the real world. The same model that scored 93% on EuroSAT detected only 19% of real deforestation — a gap caused by domain shift, since a model trained on European land failed to even recognize tropical forest as Forest.

Error taxonomy — where the 81% went

Final analysis of the misses: in all 51 missed patches, the model failed to label the forest as Forest in 2018 — 0 out of 51 were recognized correctly. This means every miss was structurally guaranteed: the change rule requires “was Forest” as its first condition, so once the model mislabeled the forest, the miss was locked in before the 2024 image was even examined. One cause — the domain gap at the recognition step — repeated 51 times, fully explains the 0.19 recall.

The 8 false alarms show the opposite pattern: all 8 were correctly labeled Forest in 2018, and several show visible partial clearing in the 2024 photos.



Figure 6: The eight false-alarm patches — all correctly labeled Forest in 2018, several with visible partial clearing by 2024.

This suggests the false alarms are largely not model errors but disagreements with Hansen's $\geq 10\%$ loss threshold or definition — cases where the model may even be detecting real change below Hansen's cutoff. In short: every failure of coverage traces to domain shift at the recognition step, while the model's positive claims were mostly defensible.

Limitations

This study covers one 11×11 km region and one pair of years, so the exact numbers may differ elsewhere. Our patches are 640 m while Hansen's pixels are 30 m, so partial clearings inside a mostly-green patch can be counted by Hansen but invisible to our rule, and the 10% loss threshold is our own choice. The Hansen year-stamps and our composite windows do not align perfectly, so some late-2018 loss is excluded from grading despite being visible to the model. We used RGB bands only — the full 13-band spectral data might reduce the domain gap. Hansen itself is a satellite-derived model, not perfect truth, and it measures tree-cover loss rather than legal deforestation. Finally, our two composites drew on different numbers of source images (12 vs 28), a small asymmetry between the before and after photos.

Conclusion

We trained a transfer-learned ResNet50 to 93% accuracy on the EuroSAT benchmark (baseline CNN: 63%), deployed it on real Sentinel-2 imagery of Rondônia, Brazil (2018 vs 2024), and graded its deforestation detections against the Hansen Global Forest Change record. On arrival in the Amazon, the model's mean confidence halved ($0.95 \rightarrow 0.50$), it mislabeled dense tropical forest as cropland and water, and its detection collapsed to precision 0.60 with recall 0.19 — it missed 81% of the real forest loss. The error taxonomy shows one cause behind every single miss: the domain gap prevented the model from recognizing tropical forest in the first place, so the change rule could never fire. The answer to our research question — a model that scores 93% in the lab, how much does it really score in the real world? — is 19%, and the difference is lost at the very first step of recognition. Benchmark accuracy is not deployment accuracy, and the gap can be almost total when the training world and the real world do not match.

Future Work

Fine-tune the model on a few hundred labeled tropical patches and measure how much the recall recovers; repeat with the 13-band spectral version of EuroSAT; test more regions and year pairs; and try a smaller patch size to catch partial clearings.

Acknowledgment

AI assistance was used for debugging code, verification of results, and language editing. Datasets: EuroSAT (Helber et al., 2019), Hansen Global Forest Change (Hansen et al., 2013), Sentinel-2 imagery via Google Earth Engine. All experiments, decisions, and findings are the author's own.